Genomic Analysis of the Rice's Whale

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Abstract

The Rice's whale, *Balaenoptera ricei*, is a recently recognized and critically endangered baleen whale species endemic to the northern Gulf of Mexico, with an estimated population of fewer than 100 individuals. Given its highly restricted distribution, limited population size, severe anthropogenic threats, and historically unresolved taxonomy within the Bryde's whale complex, a comprehensive understanding of the species' genetic landscape is essential for informed conservation. While previous studies have established species-level distinctions based on mitochondrial and morphological data, broader genomic analyses remain limited. Little is known about the Rice's whale's genome-wide variation, evolutionary history, and capacity for recovery. This ongoing study aims to characterize the genetic diversity, ancestry, and evolutionary history of the Rice's whale using whole-genome sequencing and comparative genomics. Genomic mapping and alignment of the Rice's whales' genomes and reference genomes were performed, followed by variant calling and single-nucleotide polymorphism (SNP) analysis to classify differences between the genomes. Measures of genetic distance, Dxy, were calculated, and large-scale genomic alignments were visualized via MUMmer's nucmer. Additionally, ADMIXTURE analysis was conducted to infer ancestral proportions among Rice's whales and related species. Preliminary ADMXITURE findings indicate reduced genetic diversity and inbreeding depression within the Rice's whale population, and genetic divergences suggest introgression from certain outgroup species may have occurred. However, as this project remains in progress, further analyses are ongoing to refine these patterns and explore additional aspects of population structure and demographic history. This work represents an important step toward generating a comprehensive genomic framework for *B. ricei* and will ultimately contribute to the development of conservation strategies for this threatened species.

1. Introduction

The Rice's whale, *Balaenoptera ricei*, is a newly recognized species of baleen whale found only in the northern region of the Gulf of Mexico. Originally thought to be a subspecies of the Bryde's whale, the Rice's whale was officially recognized as a distinct species in 2021 (Rosel et al., 2021). The *Balaenoptera* taxonomy has become muddled due to the Bryde's-like whales being long grouped as cryptic species, which are difficult to differentiate due to similar external appearances and overlapping ranges. There has been a history of debate on whether the Eden's whale (*B. edeni edeni,* found in the nearshore waters of the western Pacific and the Indian Ocean) and the Bryde's whale (*B. edeni brydei,* found primarily in the open oceans of the Atlantic, Pacific, and Indian Oceans) were separate species or subspecies, however, most scientists today agree that they are monotypic (belonging to the same species) (National Oceanic and Atmospheric Administration Fisheries, 2024).

The Gulf of Mexico Bryde's whale, later renamed the Rice's whale, has been grouped into the Bryde's whale species complex since its discovery by zoologist Dale W. Rice in 1965 (Marine Mammal Commission, 2025). A combination of genetic and morphological evidence urged scientists to distinguish the Rice's whale from other Bryde's-like whales, which include *B. edeni edeni* (Eden's whale), *B. edeni brydei* (Bryde's whale), and *B. omurai* (Omura's whale). When scientists analyzed the whales' mitochondrial DNA, they discovered that the Gulf whales had a distinct genetic signature not seen in other known species of baleen whales (Garrison et al., 2024; Rosel et al., 2021). This genetic analysis also revealed that the Rice's whale's closest baleen relative was the Eden's whale, as shown in Figure 1 (Rosel et al., 2021). Furthermore, in 2019, a Gulf of Mexico Bryde's whale became stranded on shore, allowing researchers to examine a complete skull, which has now been designated as the holotype of the Rice's whale (Garrison et al., 2024; Rosel et al., 2021). Skull features, particularly at the vertex (top-rear), showed unique characteristics not found in the other Bryde's whale subspecies. The frontal bones wrapped around and separated the posterior ends of the nasals in a way not seen in other species, and other traits like the narrow exposure of the frontal bones and the shape of the premaxilla distinguished them further (Rosel et al., 2021).

The Rice's whale is the only resident baleen whale to be uniquely endemic to the Gulf of Mexico, making its ecological and genetic status both distinctive and highly vulnerable. With an estimated population of fewer than 100 individuals, the Rice's whale is one of the most endangered marine mammals in the world. Its critically low numbers, restricted geographical range along the northeastern Gulf shelf, and subjection to anthropogenic threats—such as ship strikes, noise pollution, and oil exploration—have heightened the urgency of conservation efforts aimed at ensuring its survival (NOAA Fisheries, 2025). A particularly pressing concern in small, isolated populations is their low genetic diversity. Limited allele and gene variety can reduce a species' resilience to environmental changes and disease, and increase the risk of inbreeding depression, the loss of potentially adaptive genetic diversity, and the accumulation of deleterious (harmful) mutations (Rosel et al., 2021). Low genetic diversity in an already severely endangered species can quickly lead to extinction. For the Rice's whale, understanding the extent and implications of its genetic variability is fundamental for informing effective conservation strategies. While genetic research has been done to differentiate the Rice's whale from other baleen whales, little research focuses on assessing the whale's genetic history with the intent of determining their capacity for recovery. Insights into the Rice's whale's genomic landscape can help evaluate the species' capacity for adaptation and recovery and guide the implementation of tailored protective measures.

Building on previous studies that have differentiated the Rice's whale from others in the Bryde's whale complex, in this work we aim to explore the genetic architecture of the Rice's whale through comprehensive genomic analysis. The overarching goal is to better understand the evolutionary and demographic history of this species, as well as its potential for recovery in the face of severe threats and population constraints. My research, in collaboration with the Lohmuller Laboratory, is centered around several key objectives. We assess the genetic diversity of the Rice's whale by analyzing whole-genome sequencing data, compare the Rice's whale genome with those of closely related baleen whale species to contextualize genetic variation, use genomic tools to infer historical population dynamics and evolutionary relationships, identify patterns of mutation—both synonymous (tolerated) and nonsynonymous (deleterious), and assess overall genetic ancestry.

To accomplish these goals, the project is structured around several bioinformatics workflows. Genomic mapping and alignment were performed using the Burrows-Wheeler Aligner - Maximal Exact Match (BWA-MEM), a widely recognized and accurate alignment program suitable for handling complex mammalian genomes (Li, 2013). Variant calling and single-nucleotide polymorphism (SNP) analysis were conducted to identify genetic differences within and between species, with a focus on characterizing the functional consequences of mutations. Measures of genetic distance (dxy) were calculated to assess evolutionary divergence, employing the University of California, Santa Cruz (UCSC) LiftOver data to efficiently translate genome coordinates across species (Perez et al., 2025). Additionally, genome-to-genome alignments were generated using nucmer, a component of the MUMmer system, to visualize large-scale structural similarities and differences between genomes (Marçais et al., 2018). Finally, we used Admixture to estimate the Rice's whales' global ancestry, the proportion of each individual's genome that comes from each ancestral population (Alexander et al., 2009). These combined approaches were selected for their accuracy, computational efficiency, and established utility in comparative genomics. Ultimately, this work seeks not only to advance our understanding of *Balaenoptera ricei* at the genetic level but also to provide data-driven evidence that can influence conservation policy and action. By highlighting the whale's genetic strengths and vulnerabilities, this genomic investigation contributes to the broader goal of preserving one of the ocean's most imperiled species.

2. Methods

2.1 Genome-to-Genome Alignment

In the first step of our project, we obtained our reference genomes and the Rice's whale genomes from our NOAA collaborators. To begin the preprocessing for our comparative analysis, we mapped and aligned the Rice's whale genome to our reference genomes of other whale species (Bryde's, Sei, Fin, Minke, and Blue whales). Mapping and alignment allow us to identify where DNA sequences of the Rice's whale genome align within a reference genome. We chose to accomplish this by using BWA-MEM, as it is one of the most accurate alignment programs available (Li, 2013). As highlighted by Nobrega and Pennacchio (2003), comparative genomic analysis allows researchers to explore evolutionary relationships by identifying conserved sequences that provide insights into species divergence. In the case of the Rice's whale, our research seeks to compare its genome with that of closely related baleen whale species to better understand its evolutionary history and genomic distinctiveness. Our Rice's whale genomes were aligned to the Bryde's whale and Sei whale references using BWA-MEM2 on our Hoffman2 cluster. The references were first indexed using the BWA-MEM2 vcftools index command, then paired-end reads were mapped with and piped directly into the Picard AddOrReplaceReadGroups function, which sorts the alignments by coordinate and adds read-group tags, returning binary alignment map (BAM) files.

2.2 Single Nucleotide Polymorphism Analysis

We then performed variant calling and single-nucleotide polymorphism (SNP) analysis on the generated BAM files by utilizing a combination of bcftools functions (Danecek et al., 2021). We used bcftools mpileup to compute genotype likelihoods, piped into bcftools call to produce a compressed variant call format file (VCF) of all sites, and indexed the result. SNP analysis involves examining variations at single positions in the DNA sequence among individuals. SNPs are the most common type of genetic variation, where a single nucleotide (A, T, C, or G) is substituted for another at a specific location in the genome (Lee & Lee, 2000). SNPs reveal genetic differences across species and allow us to determine individual mutations in the whale genomes. We classified these mutations based on their functional consequences: synonymous vs. nonsynonymous mutations and tolerated vs. deleterious mutations. Synonymous mutations, although they change which allele is present in the genome, do not affect which amino acid is coded. Nonsynonymous mutations, however, mutate the DNA sequence in a way that changes the amino acid that is produced. Generally, we classify synonymous mutations as tolerated, meaning that they do not significantly affect an individual's function, and nonsynonymous mutations as deleterious, those that do have an effect on function. To begin our SNP analysis, all VCFs were divided into 1 Mb windows to quantify genotype counts across the genome in a way that allows us to observe patterns across the entire genome. We extracted each chromosome’s length from the reference scaffold index and used Bedtools makewindows to generate 1 Mb intervals (Quinlan and Hall, 2010). For each window, we parsed the corresponding VCF with an awk filter command, a standard linux text processing function that parses data line-by-line, to pull sites within the interval, then tallied sample genotypes–heterozygous (0/1), homozygous reference (0/0), homozygous alternate (0/1), or missing (./.)–via an awk‑based counter.

2.3 Whole Genome Alignment Visualizations

Next, we generated genome-to-genome alignment visuals using nucmer, the sequence alignment component of the MUMmer system. These programs are primarily used for aligning whole genomes to each other, as they are capable of detecting sequence identity, rearrangements, duplications, and unaligned regions (Marçais et al., 2018). This allowed us to compare the entire Rice’s whale genome with those of our outgroup species, instead of just looking at individual genome segments. We performed genome‑to‑genome alignments between the Rice’s whale assembly and both the Bryde’s whale and sei‑whale references using MUMmer4. First, each pair of assemblies was aligned with nucmer. The raw outputs were then filtered for one‑to‑one alignments of ≥10 kb and ≥99% identity using delta-filter, and these mappings were visualized with mummerplot.

2.4 Genetic Divergence Calculations

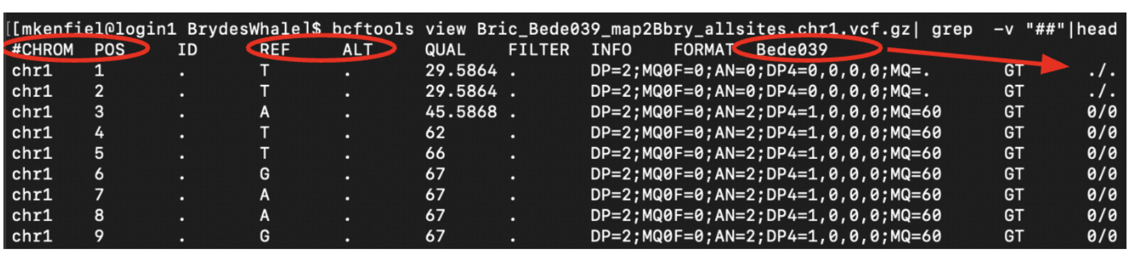
Our study also involved calculating genetic distance (dxy), a measure of nucleotide divergence between two species. Dxy allows us to approximate how closely related the Rice’s whale and outgroup species are by finding functionally conserved sequences of the two genomes and determining the evolutionary distance between them (Nobrega & Pennacchio, 2004). We decided to run our dxy calculations using liftover data generated from the UCSC LiftOver program. LiftOver analysis for comparative genomics is a method used to convert genome coordinates (positions of genes or variants) from one reference genome to another without re-aligning the raw sequence data (Luu et al., 2020; Perez et al., 2025). It allows for cross-species mapping in a more computationally efficient manner. First, we concatenated the SNP count files into a single bed file of genotype counts. We then ran the UCSC liftOver function with the outgroup chain file to translate each interval's coordinates. In R, we computed Dxy for each 1 Mb segment with the following equation: $Dxy \;=\; \frac{het + 2\times alt\_homo}{2\times total\_genotypes}$. In this equation, het represents the heterozygous genotype counts, alt_homo represents the alternative (Rice's whale) homozygous genotype counts, and the total_genotypes is the total count of all genotypes, homozygous, heterozygous, and missing. The alt_homo and total_genotypes variables are multiplied by twso to account for the two alleles in each genotype. We then plotted the windowed Dxy along each chromosome using ggplot2.

2.5 ADMIXTURE Analysis

Lastly, we estimated individual ancestry proportions using the program ADMIXTURE, which implements a fast maximum-likelihood model–based clustering algorithm (Alexander et al., 2009). Input genotypes were filtered for minor-allele frequency ≥ 0.05, meaning that for each genetic variant (SNP) we keep in the analysis, the less-common allele must appear in at least 5% of all chromosomes in the dataset. This ensures we are using only variants that are sufficiently frequent to contribute a meaningful signal to the admixture model. We ran ADMIXTURE for 8 ancestral populations (K) with 10 independent replicates per K. The optimal K was chosen by minimizing five-fold cross-validation error. Finally, Q-matrices, representations of individual ancestry proportions, from the replicate with the lowest cross-validation error at the selected K were plotted as stacked barplots in R to visualize admixture patterns among Rice's whales and outgroups.

3. Results

3.1 Genome-to-Genome Alignment, Variant Calling, and SNP Analysis

a) 

```
[mkenfiel@login1 BrydesWhale]$ bcftools view Bric_Bede039_map2Bbry_allsites.chr1.vcf.gz| grep  -v "##"|head
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  Bede039
chr1    1       .       T       .       29.5864 .       DP=2;MQ0F=0;AN=0;DP4=0,0,0,0;MQ=.       GT      ./.
chr1    2       .       T       .       29.5864 .       DP=2;MQ0F=0;AN=0;DP4=0,0,0,0;MQ=.       GT      ./.
chr1    3       .       A       .       45.5868 .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    4       .       T       .       62      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    5       .       T       .       66      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    6       .       G       .       67      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    7       .       A       .       67      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    8       .       A       .       67      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
chr1    9       .       G       .       67      .       DP=2;MQ0F=0;AN=2;DP4=1,0,0,0;MQ=60      GT      0/0
```

b)

Reference Allele	Alternative Allele	Genotype	Interpretation
A	.	0/0	A/A (Homozygous Reference)
C	G	1/1	G/G (Homozygous Alternative)
T	C	0/1	T/C (Heterozygous)
G	.	./.	Missing Data

Figure 1. Example of a variant call file from the mapping of Rice's whale individual Bede039 and the Bryde's whale reference genome. **a)** First 10 lines of the file. Red markings show columns used further downstream. **b)** Table showing how to interpret the file. The reference allele corresponds with the outgroup species, while the alternate allele corresponds to the Rice's whale.

Following genome alignment of Rice's whale individuals to the Bryde's whale, Sei whale, and Eden's whale reference genomes, variant calling was successfully performed using bcftools. This process generated variant call format (VCF) files containing identified single-nucleotide polymorphisms (SNPs) for each Rice's individual's genome relative to the reference genomes. An example of the resulting VCF files is shown in Figure 1b, illustrating the structure and key columns utilized in subsequent analyses. Each VCF file records the chromosome number, genomic position, reference and alternate alleles, and genotype calls for every variant site detected. The table (Fig. 1a) explains how to interpret these VCF files, specifically showing how different SNPs are labeled as either homozygous reference, homozygous alternative, heterozygous, or missing, when genomic data has been lost in the initial sequencing steps. These variant datasets form the foundation for all downstream analyses, including SNP annotation, genetic distance (Dxy) calculations, and ADMIXTURE-based ancestry inference. At this stage, no formal variant interpretation or summary statistics were performed; rather, the primary objective was to generate genome-wide variant datasets necessary for comparative and population genomic analyses conducted in subsequent steps.

```
[[mkenfiel@login4 chromCount]$ head Bede039_Bric-Bbry_SNPcount_chr1_1Mb.txt
chr1 0 1000000  186 988018 1880 344
chr1 1000000 2000000    175 974822 1729 377
chr1 2000000 3000000    220 988881 2323 432
chr1 3000000 4000000    165 992195 1786 404
chr1 4000000 5000000    397 984711 2345 604
chr1 5000000 6000000    260 978887 2057 451
chr1 6000000 7000000    326 982450 1787 384
chr1 7000000 8000000    177 990153 2217 396
chr1 8000000 9000000    190 987604 2105 474
chr1 9000000 10000000   327 983449 3817 830
```

Figure 2. Example of an SNP counts by windows file, calculated from the variant call file between Bede039 and the Bryde's whale reference genome. From left to right, these columns represent the chromosome, window start position, window end position, and the counts for heterozygous, reference homozygous, alternative homozygous, or missing SNPs.

To quantify genome-wide patterns of genetic variation in Rice's whale individuals relative to each reference genome, single-nucleotide polymorphism (SNP) counts were counted within consecutive, non-overlapping 1 megabase (Mb) windows across all chromosomes. For each window, the number of SNP sites falling into four genotype categories was tallied based on the information recorded in the corresponding VCF files. The resulting SNP count files (illustrated in Figure 2) include, for each window, the chromosome number, window start and end positions, and the counts for each genotype category: heterozygous, homozygous for the reference allele, homozygous for the alternate allele, and missing (no genotype call). These files serve as an essential intermediate data product, providing a summary of genome-wide variant distribution patterns that support subsequent analyses, including calculations of genetic divergence between genomes, as well as broader assessments of genome-wide diversity landscapes.

3.2 Whole Genome Alignment Visualizations

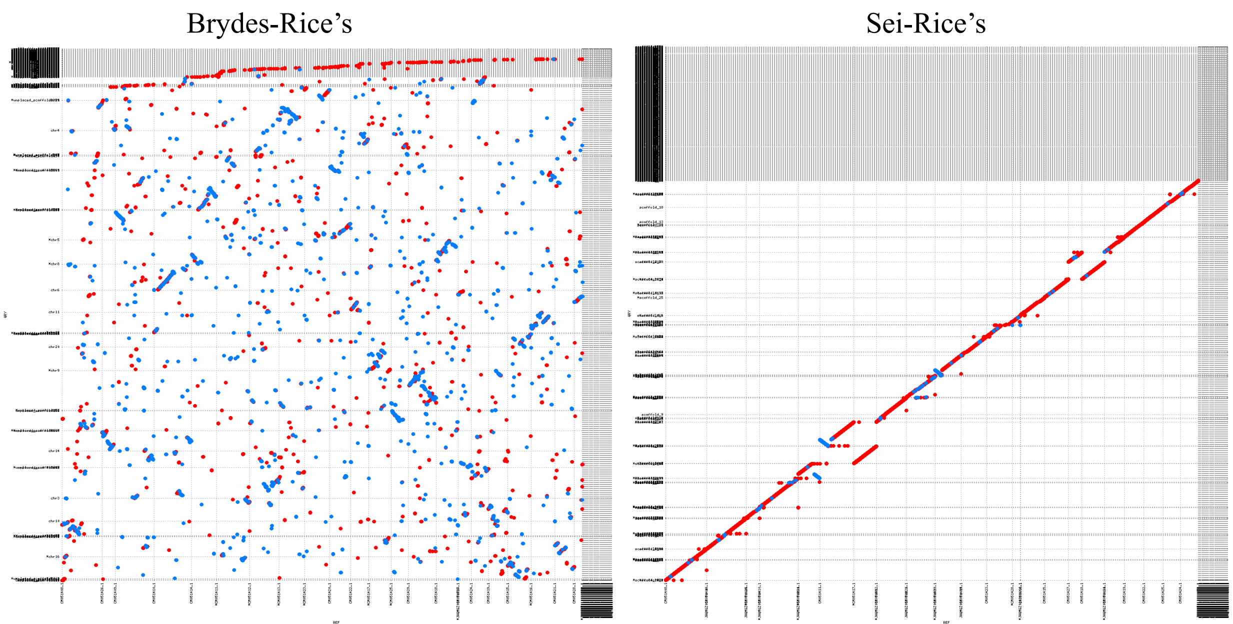
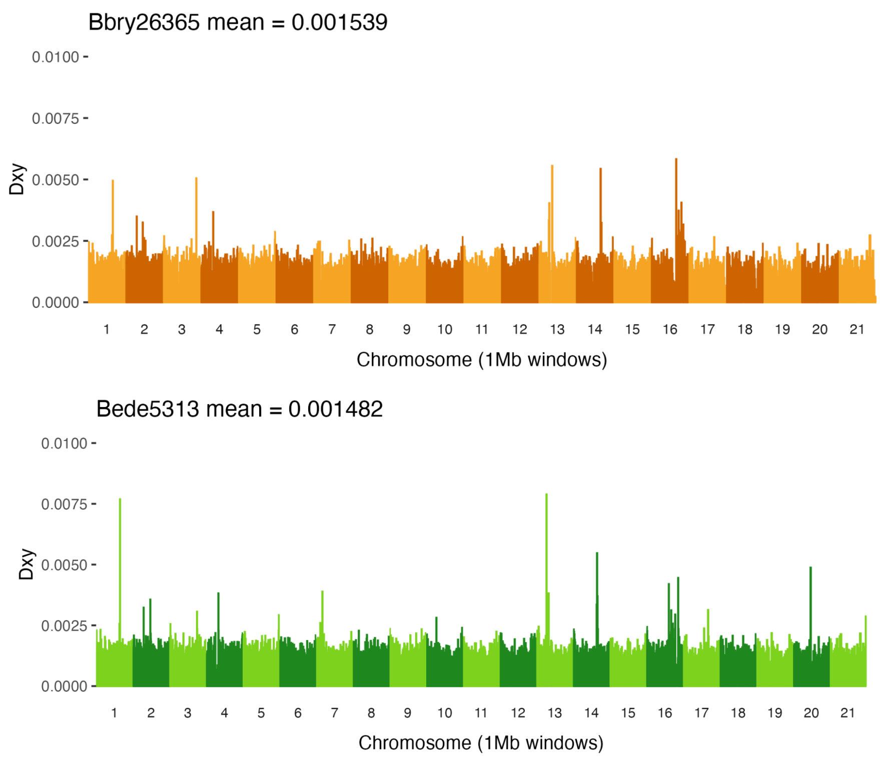


Figure 3. MUMmer plots visualizing the alignment of the Rice's whale genome with that of the Bryde's whale (left) and Sei whale (right). Red segments represent forward matches between the genomes, and blue segments represent reverse matches. Markings on the axes represent the chromosomes of the corresponding genome.

Genome-to-genome alignments between the Rice's whale genome assembly and two outgroup reference genomes, the Bryde's whale and the Sei whale, were performed using MUMmer's nucmer alignment tool and visualized with mummerplot (Figure 3). The alignment of the Rice's whale genome to the Sei whale genome (aligned by our collaborators at NOAA) produced a largely continuous, linear pattern, indicative of strong synteny and conserved structure between these two species. Only a small number of structural rearrangements were apparent, consistent with expectations given their close evolutionary relationship within the Balaenopteridae family. The uniformity of forward (red) and reverse (blue) match segments further supports the high-quality assembly and the suitability of the Sei whale genome as a reference for comparative analyses.

In contrast, alignment of the Rice's whale genome to the Bryde's whale reference genome produced a highly fragmented and disordered pattern, with matches dispersed irregularly across both axes. Since the Bryde's whale and the Rice's whale are closely related species, their genome-to-genome alignment should return a linear, relatively consistent alignment pattern, similar to the Rice's-Sei alignment map. The lack of clear linear alignment and the abundance of scattered segments suggested potential issues with the quality or completeness of the Bryde's whale reference assembly, which was sourced from the China National GeneBank Database. Based on these observations, we determined that outsourced assemblies were not suitable for downstream comparative genomic analyses. Consequently, subsequent analyses prioritized comparisons using genome assemblies generated by our NOAA collaborators, which demonstrated higher assembly quality and contiguity, ensuring more reliable and biologically meaningful comparative genomic insights.

3.3 Genetic Divergence Calculations



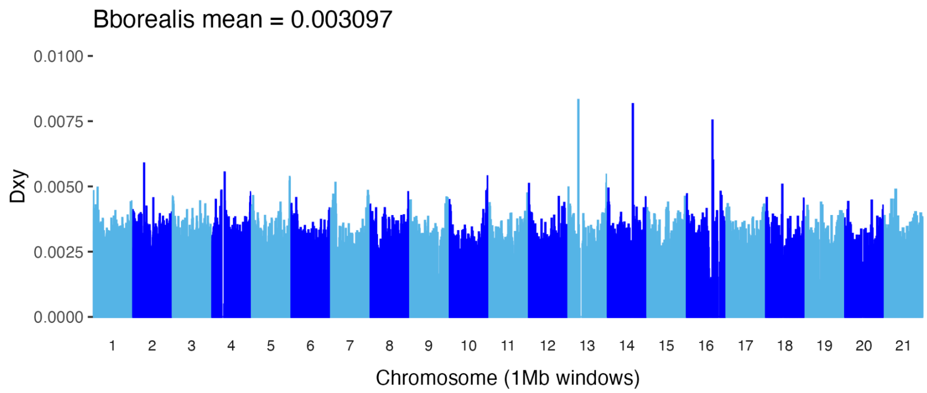
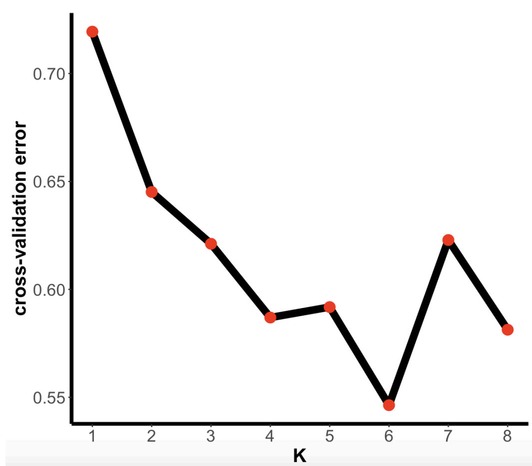


Figure 4. Plots showing genetic divergence (Dxy) between the Rice's whale and three outgroup species: Bryde's (orange), Eden's (green), and Sei (blue). Dxy was calculated and plotted for 1 Mb windows across the genome. The mean Dxy for each pairing was calculated and printed above the corresponding plot.

Dxy values were calculated in 1 Mb windows across the genome to estimate pairwise genetic divergence between the Rice's whale and three outgroup species: Bryde's whale, Eden's whale, and Sei whale (Figure 4). As expected, the Rice's whale exhibited the highest mean Dxy with the Sei whale (0.003097), consistent with its more distant phylogenetic relationship. The Rice's whale showed a slightly lower mean Dxy with Eden's whale (0.001482) compared to Bryde's whale (0.001539), aligning with established phylogenetic relationships within the Bryde's whale complex. These genome-wide divergence estimates provide a baseline for evaluating evolutionary distances and identifying regions of relative conservation and divergence across the genome.

3.4 ADMIXTURE Analysis



K	CVE
1	0.71940
2	0.64513
3	0.62113
4	0.58695
5	0.59180
6	0.54636
7	0.62287
8	0.58130

Figure 5. Cross-validation error plot to determine optimal number of clusters, K, for ADMIXTURE Analysis. The table on the right gives the exact errors.

To determine the optimal number of ancestral populations (K) for the ADMIXTURE analysis, cross-validation error was calculated for values of K ranging from 1 to 8 (Figure 5). The plot shows that the cross-validation error tends to decrease with increasing K (with an outlier of K=7), reaching its lowest values at K=6 (0.54636) and K=8 (0.58130). These values were selected as the optimal Ks for the following ancestry proportion analyses, as they minimize cross-validation error while balancing model complexity.

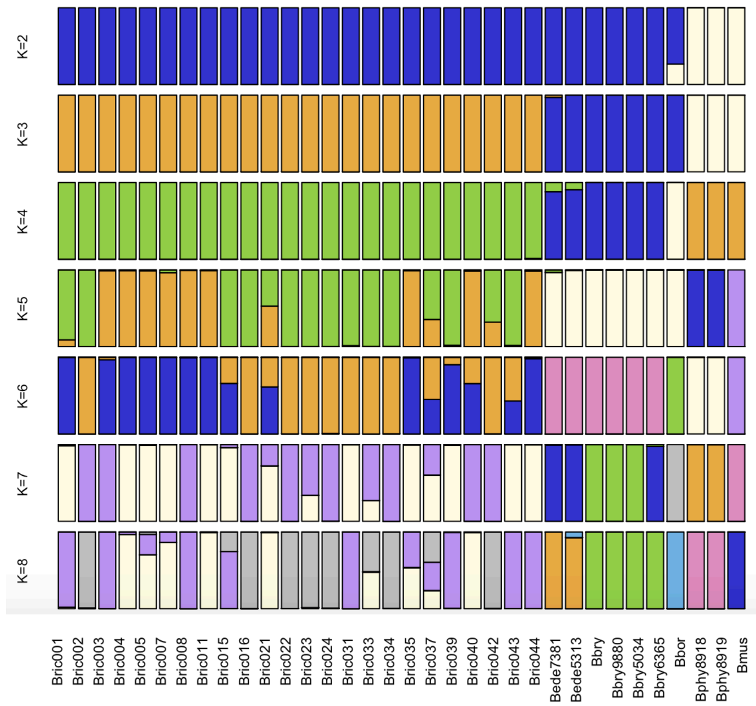


Figure 6. ADMIXTURE plot of K values spanning from 2-10, with bars corresponding to individual whales. From left to right: the first 24 bars represent Rice's whale individuals, followed by 2 Eden's whales, 4 Bryde's whales, 1 Sei whale, 2 fin whales, and 1 blue whale.

Ancestry proportions for each individual were estimated using ADMIXTURE across K values ranging from 1 to 10 (Figure 6). As expected, at lower K values, individuals from different species share broad ancestry components, reflecting deeper phylogenetic relationships. As K increases, the analysis begins to resolve finer-scale population structure and species-level distinctions. By K=6, most species are beginning to form distinct clusters, and by K=8, clear separation is observed between each species, including within the Rice's whale population.

Within the Rice's whale group, ADMIXTURE revealed noticeable variation in ancestry proportions among individuals. Some individuals display shared ancestry components not present in others, suggesting either close familial relationships (such as parent-offspring or sibling pairs) or signals of inbreeding within this small, isolated population. Given the Rice's whale's extremely limited population size and restricted geographic range, such patterns of relatedness are not unexpected and are consistent with what would be predicted in cases of inbreeding depression or limited gene flow.

4. Discussion

4.1 Genetic Divergence

Our analysis of genome-wide genetic divergence (Dxy) between the Rice's whale and the three outgroup species Bryde's whale, Eden's whale, and Sei whale, offers new insights into the evolutionary relationships within the *Balaenopteridae* family. The patterns we observed are broadly consistent with established phylogenetic relationships but also suggest potential introgression (interbreeding between differing species) that warrants further exploration.

The low mean Dxy observed between the Rice's whale and the Eden's whale is likely due to their close phylogenetic affinity. As expected, the divergence values between these two species were lower than those between the other outgroup species, as the Rice's whale is more closely related to the Eden's whale. These whales are expected to share conserved genomic regions due to their recent common ancestry, therefore, the reduced Dxy here likely reflects phylogenetic proximity rather than introgression. The relatively low mean Dxy between the Rice's whale and Bryde's whale, on the other hand, suggests a close genetic relationship, which may be partially explained by historical introgression. Since the Rice's whale is less related to the Bryde's whale, it is interesting that the mean Dxy values between the two outgroup species are so similar. Our results support the possibility that Rice's whale genomes harbor introgressed regions from Bryde's whales. These regions of reduced divergence could reflect segments of shared ancestry resulting from past interbreeding events; however, further research comparing Dxy to heterozygosity needs to be done to definitively conclude this idea. In contrast to the Eden's and Bryde's whales, the highest mean Dxy values were observed between the Rice's whale and the Sei whale, supporting their more distant evolutionary relationship within *Balaenopteridae*. The absence of unusually low divergence windows in this comparison suggests that there has been limited or no recent introgression between these lineages.

Taken together, these divergence estimates contribute to our understanding of the Rice's whale's genomic history and its evolutionary relationships with closely related species. While genome-wide Dxy patterns provide a useful baseline, further analysis will be essential to fully characterize the extent of introgression events and to understand their potential influence on the conservation of the Rice's whale.

4.2 ADMIXTURE Analysis

The ADMIXTURE analysis provided additional resolution into the population structure and ancestry composition of the Rice's whale relative to closely related outgroup species. Our results demonstrated clear genetic distinctions between species-level clusters, while also uncovering evidence of close familial relationships and potential inbreeding within the Rice's whale population.

Across a range of K values, the ADMIXTURE results consistently recovered distinct ancestry components corresponding to the recognized species in our dataset. By K=6, species-level clustering was largely resolved, and by K=8, each species formed discrete genetic clusters. These patterns support the current taxonomic separation of the Rice's whale from other members of the Bryde's whale complex, reinforcing the decision to formally reclassify it as a distinct species. The clear separation observed in our ADMIXTURE plots provides independent, genome-wide evidence of the Rice's whale's genetic distinctiveness, consistent with prior phylogenetic and morphological studies.

Within the Rice's whale population, ADMIXTURE revealed notable heterogeneity in individual ancestry proportions. While the population as a whole formed a distinct genetic cluster, several individuals displayed further clustering, which provides insight into relatedness among the Ruce's whale individuals. This clustering could be the result of direct familial relationships, such as parent-offspring pairs or siblings, or it may indicate signs of inbreeding, a concern for small, isolated populations. Inbreeding depression, characterized by the reduced fitness of offspring as a result of genetically similar individuals breeding, is a plausible outcome in such contexts. The variability in individual ancestry proportions seen in our ADMIXTURE analysis aligns with the demographic history of the Rice's whale, which is known to have an extremely small population size and a restricted geographic distribution.

4. Conclusion

Our study provides new insights into the genomic history and population structure of the critically endangered Rice's whale. Through genetic divergence (Dxy) analyses, we identified evidence suggesting historical introgression between the Rice's whale and the Bryde's whale, as reflected by their relatively low mean genetic distance. In contrast, while the Rice's whale also exhibited low Dxy values with the Eden's whale, this pattern is more likely attributable to their close phylogenetic relationship rather than evidence of interbreeding. The higher divergence observed between the Rice's whale and the Sei whale further supports the absence of significant gene flow between these lineages.

Complementing these findings, our ADMIXTURE analysis revealed clear species-level classifications among all sampled individuals, providing strong support for the continued recognition of the Rice's whale as a distinct species within the Bryde's whale complex. Within the Rice's whale population, patterns of shared ancestry components suggest genetic relatedness among individuals, likely due to the species' small population size, restricted range, and limited gene flow, which raises concerns about the presence of inbreeding depression.

While this project remains ongoing, our current findings offer a valuable perspective on the genetic architecture and demographic history of the Rice's whale. The evidence of inbreeding and historical introgression underscores the species' vulnerability and highlights the urgent need for targeted conservation measures. Future work should focus on more detailed analyses of relatedness, genetic diversity, and demographic history to better inform conservation strategies aimed at preserving the genetic health and long-term viability of this unique species.

Ackoweldgements

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